

Xinyue Liang

Research Associate

Department of Mechanical Engineering
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Education

- 2022.08 - 2023.05 **BEng in Department of Industrial Engineering (Transferred)** ,
Department of Industrial Engineering, Tsinghua University, Beijing, China
○ GPA: **3.94/4.0**.
- 2023.05 - 2026.06 **BEng in Computer Science and Technology (Yao Class)** ,
Institute for Interdisciplinary Information Sciences, Tsinghua University, Beijing, China
○ Relevant GPA: **3.8/4.0** for all courses, including Advanced Computer Graphics, Theory of Computation, Machine Learning, Algorithm Design, Abstract Algebra.

Publication

- In preparation **DINO-ECHO: A Foundation Model for Echocardiography Report Generation**,
In preparation for submission to Nature Medicine
○ **Xinyue Liang***, Hanwen Xu*, Zhiping Xiao, Shin Lin, Sheng Wang
○ *Co-first authors (Xinyue Liang listed first)
○ Code available at <https://github.com/xy-liang22/DINO-ECHO>.

Research Experience

- 2026.02 - Present **A Self-Reflective Iterative Framework for Ultrasound Video Understanding**,
Department of Mechanical Engineering, University of Hong Kong,
Advisor: Prof. Zhongliang Jiang
○ Developed a self-reflective iterative framework with CoTT techniques to enhance LLMs' ability to extract temporal information and clinical details from ultrasound videos.
○ This work formed my undergraduate thesis.
- 2025.03 - 2026.01 **DINO-ECHO: A Foundation Model for Echocardiography Report Generation**,
Paul G. Allen School of Computer Science & Engineering, University of Washington,
Advisor: Prof. Sheng Wang
○ Developed multi-modal models for echocardiographic video analysis and clinical report generation.
○ Contributed to a co-first-authored paper planned for submission to *Nature Medicine*.
○ The project can be found at <https://github.com/xy-liang22/DINO-ECHO>.
- 2024.10 - 2024.12 **LLMs in Naming Relations in Chinese with Chain of Thoughts**,
IIIS, Tsinghua University
○ Improved MiniCPM for Chinese relation extraction via chain-of-thought fine-tuning.
○ The project can be found at <https://github.com/xy-liang22/NLP-Project>.

Awards and Scholarships

- 2025 **Literature and Art Scholarship**, *Tsinghua University*
2024 **Literature and Art Scholarship**, *Tsinghua University*
2023 **Merit Student Scholarship (Top 5)**, *Tsinghua University*
2021 **Silver Medal**, *Chinese Mathematical Olympiad (CMO)*

Extracurricular Activities

2024.03 - Present **Member of Tsinghua University Student Chorus**
2024.07 - 2024.08 **Volunteer Teaching in Shigatse, Tibet, China**
2023.07 - 2023.08 **Volunteer Teaching in Ledong, Hainan, China**
2022.10 - 2025.11 **Directing programs in Student Festival**

Technical Skills

Languages Python, C++
Frameworks PyTorch
Tools LaTeX, Git